

2. Create a scene, where a candle is lit (with shadows and smoke), on mouse click using Unity.

STEP 1: Create Candle Object

Candle Body

1. **Right-click Hierarchy → 3D Object → Cylinder**
2. Rename → `Candle`
3. Adjust Transform:
 - Scale:

```
X: 0.3  
Y: 1  
Z: 0.3
```

Wick

1. Right-click → **3D Object → Cylinder**
2. Rename → `Wick`
3. Set:

```
Scale: (0.05, 0.2, 0.05)  
Position Y: top of candle (~1.1)
```

STEP 2: Create Flame

Option A (Simple – Recommended)

1. Right-click → **Effects → Particle System**
 2. Rename → `Flame`
 3. Move it to wick position
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Flame Settings (IMPORTANT)

Select Flame → Inspector:

Main:

- Duration: 1
- Looping ✓
- Start Lifetime: 0.3
- Start Speed: 0.1
- Start Size: 0.2
- Start Color: Orange/Yellow

Emission:

- Rate over Time: 30

Shape:

- Cone
- Angle: 10

Renderer:

- Material → Default-Particle or Fire-like

💡 STEP 3: Add Light (VERY IMPORTANT)

1. Right-click → **Light** → **Point Light**
2. Rename → `CandleLight`
3. Position at flame

Settings:

- Range: 3-5
- Intensity: 2-4
- Color: Warm Yellow/Orange
- Shadows: ✓ Enable

☞ This gives **real candle glow + shadows**

☞ STEP 4: Add Smoke

1. Duplicate Flame:
→ Rename → Smoke

Change settings:

- Start Color → Gray
- Start Size → 0.3
- Start Speed → 0.5
- Lifetime → 1.5
- Emission → 10

☞ Smoke should rise slowly

□ STEP 5: Organize Hierarchy

Make this structure:

```
Candle
├── Wick
├── Flame
├── Smoke
└── CandleLight
```

☞ Drag all inside Candle

📖 STEP 6: Create Script

1. Right-click → Create → C# Script
2. Name → CandleToggle

Paste this:

```
using UnityEngine;
using UnityEngine.InputSystem;

public class CandleToggle : MonoBehaviour
{
    public ParticleSystem flame;
    public ParticleSystem smoke;
    public Light candleLight;

    private bool isLit = true;

    void Update()
    {
        if (Mouse.current.leftButton.wasPressedThisFrame)
        {
```

```
        ToggleCandle();
    }
}

void ToggleCandle()
{
    if (isLit)
    {
        flame.Stop();
        smoke.Stop();
        candleLight.enabled = false;
    }
    else
    {
        flame.Play();
        smoke.Play();
        candleLight.enabled = true;
    }

    isLit = !isLit;
}
}
```

🔗 STEP 7: Attach Script

1. Select **Candle object**
 2. Drag script onto it
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🔑 STEP 8: Assign Components

In Inspector:

- Flame → drag Flame
 - Smoke → drag Smoke
 - Candle Light → drag CandleLight
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● STEP 9: Enable Shadows

Select **CandleLight**:

- Shadows → **Soft Shadows**

Also:

- Make sure ground exists:

- Right-click → 3D Object → Plane
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► **STEP 10: Run**

1. Click **Play**
2. Click mouse

☞ Candle should:

- Turn OFF (flame + light gone)
- Turn ON again on next click